Limits, Continuity, Differentiability — Revision Summary

Sequences, series, function limits, and the calculus of one variable

1 Sequences

A sequence (a_n) converges to L if for every $\varepsilon > 0$ there exists N such that $|a_n - L| < \varepsilon$ for all $n \geq N$. Limits are unique when they exist.

1.1 Core convergence theorems

- **Monotone Convergence:** A monotone sequence converges iff it is bounded. (Bounded above + increasing $\Rightarrow$ converges to its supremum.)
- **Bolzano–Weierstrass:** Every bounded sequence has a convergent subsequence.
- **Cauchy criterion:** (a_n) converges iff for every $\varepsilon > 0$ there exists N with $|a_m - a_n| < \varepsilon$ for all $m, n \geq N$. (Lets you prove convergence without knowing the limit.)
- **Algebra of limits:** If $a_n \to a$ and $b_n \to b$, then $a_n \pm b_n \to a \pm b$, $a_n b_n \to ab$, $a_n/b_n \to a/b$ (if $b \neq 0$).

1.2 Squeeze (sandwich) theorem

Theorem (Squeeze)**.** *If $a_n \leq b_n \leq c_n$ eventually and $a_n, c_n \to L$, then $b_n \to L$.*

When to use. The target sequence is messy but trapped between two clean ones. Workhorse for sequences involving $\sin, \cos$, fractional parts, or anything bounded multiplied by something tending to zero.

Standard examples. $\dfrac{\sin n}{n} \to 0$ (squeeze between $\pm\dfrac{1}{n}$); $\dfrac{n!}{n^n} \to 0$ (squeeze using $\dfrac{n!}{n^n} \leq \dfrac{1}{n}$).

1.3 Stolz–Cesàro theorem (the discrete L'Hôpital)

Theorem (Stolz–Cesàro)**.** *Let (b_n) be strictly monotone and unbounded. If $\lim_{n\to\infty} \dfrac{a_{n+1} - a_n}{b_{n+1} - b_n} = L$, then $\lim_{n\to\infty} \dfrac{a_n}{b_n} = L$.*

When to use. Sequence ratios where the numerator or denominator is a sum, a product, or otherwise built recursively. The discrete analogue of L'Hôpital: replace differences by ratios of consecutive differences.

Canonical applications.

- $\lim_{n\to\infty} \dfrac{1 + 2 + \cdots + n}{n^2} = \lim \dfrac{n+1}{(n+1)^2 - n^2} = \dfrac{1}{2}$.
- $\lim_{n\to\infty} \dfrac{1^k + 2^k + \cdots + n^k}{n^{k+1}} = \dfrac{1}{k+1}$ for any $k > -1$.
- Cesàro mean: if $a_n \to L$, then $\dfrac{a_1 + \cdots + a_n}{n} \to L$. (Set $a_n = a_n$, $b_n = n$.)
- If $a_n > 0$ and $a_{n+1}/a_n \to L$, then $a_n^{1/n} \to L$. (Take logs and apply Cesàro mean.)

1.4 Stirling's approximation — the MCQ shortcut

For large n:

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n, \quad \text{or simply} \quad n! \approx n^n e^{-n} \quad \text{(when only the exponential order matters).}$$

The cruder form $n! \approx (n/e)^n$ is enough for limits involving nth roots — the polynomial factor $\sqrt{2\pi n}$ vanishes after taking nth roots.

Killer application: central binomial coefficient.

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} \sim \frac{(2n/e)^{2n}\sqrt{4\pi n}}{[(n/e)^n\sqrt{2\pi n}]^2} = \frac{4^n}{\sqrt{\pi n}}, \quad \text{so} \quad \binom{2n}{n}^{1/n} \to 4.$$

More applications.

- $\lim_{n\to\infty} \dfrac{(n!)^{1/n}}{n} = \dfrac{1}{e}$. (Stirling: $(n!)^{1/n} \sim n/e$.)
- $\lim_{n\to\infty} \binom{kn}{n}^{1/n} = \dfrac{k^k}{(k-1)^{k-1}}$ for any $k \geq 2$.

- $\lim_{n\to\infty} \frac{n^n}{e^n n!} = 0$ (more precisely, $\sim \frac{1}{\sqrt{2\pi n}}$, which $\to 0$).

1.5 Rational and irrational sequential limits

Density tricks. Both $\mathbb{Q}$ and $\mathbb{R}\setminus\mathbb{Q}$ are dense in $\mathbb{R}$. So for any $x \in \mathbb{R}$, there exist sequences $(q_n) \subset \mathbb{Q}$ and $(\alpha_n) \subset \mathbb{R}\setminus\mathbb{Q}$ with $q_n \to x$ and $\alpha_n \to x$. This is the engine behind "two-sequence" arguments for non-existence of limits.

Classic non-existence. Define $f(x) = 1$ if $x \in \mathbb{Q}$, 0 otherwise (the Dirichlet function). For any x, pick rational $q_n \to x$ and irrational $\alpha_n \to x$: $f(q_n) \to 1$, $f(\alpha_n) \to 0$. So $\lim f$ exists nowhere, and f is discontinuous everywhere.

No function is continuous *only* at the rationals. The set of points of continuity of any function $f : \mathbb{R} \to \mathbb{R}$ is a G_δ set (a countable intersection of open sets). The rationals are *not* a G_δ set in $\mathbb{R}$ (Baire category), so no f exists with continuity set exactly $\mathbb{Q}$. Conversely, the Thomae (popcorn) function $f(p/q) = 1/q$, $f(\text{irrational}) = 0$ is continuous *exactly at the irrationals* — this is possible because $\mathbb{R}\setminus\mathbb{Q}$ *is* a G_δ.

2 Series

A series $\sum a_n$ converges if its partial sums $S_n = a_1 + \cdots + a_n$ converge.

2.1 Convergence tests

Test	Statement / when to use
nth-term test	If $a_n \not\to 0$, the series diverges. (Necessary, not sufficient.)
Geometric series	$\sum r^n$ converges iff $\lvert r\rvert < 1$, to $1/(1-r)$.
p-series	$\sum 1/n^p$ converges iff $p > 1$.
Comparison	$0 \le a_n \le b_n$: $\sum b_n$ converges $\Rightarrow \sum a_n$ converges.
Limit comparison	$\lim a_n/b_n = L \in (0,\infty)$: both converge or both diverge.
Ratio test	$L = \lim \left\lvert \frac{a_{n+1}}{a_n} \right\rvert$: $L < 1$ converges, $L > 1$ diverges, $L = 1$ inconclusive.
Root test	$L = \lim \lvert a_n\rvert^{1/n}$: same trichotomy. Stronger than ratio test in general.
Integral test	f positive, decreasing, $a_n = f(n)$: $\sum a_n$ and $\int_1^\infty f$ both converge or both diverge.
Alternating (Leibniz)	$\sum (-1)^n a_n$ converges if $a_n \downarrow 0$.
Dirichlet/Abel	$\sum a_n b_n$ converges if partial sums of a_n are bounded and $b_n \downarrow 0$ (Dirichlet); or if $\sum a_n$ converges and b_n is monotone bounded (Abel).

2.2 Absolute vs conditional convergence

- $\sum \lvert a_n\rvert$ converges $\Rightarrow \sum a_n$ converges (*absolute* convergence).
- Absolute convergence allows arbitrary rearrangement; conditionally convergent series can be rearranged to sum to *any* real number (Riemann rearrangement theorem).
- Example: $\sum (-1)^{n+1}/n = \ln 2$ converges conditionally; $\sum 1/n$ diverges.

2.3 Telescoping and partial fractions

If $a_n = b_n - b_{n+1}$, then $\sum_{n=1}^N a_n = b_1 - b_{N+1}$. Standard moves: $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$, so $\sum 1/[n(n+1)] = 1$.

3 Limits of functions

Theorem (ε-δ definition)**.** $\lim_{x\to a} f(x) = L$ *iff for every* $\varepsilon > 0$ *there exists* $\delta > 0$ *such that* $0 < \lvert x-a\rvert < \delta$ *implies* $\lvert f(x) - L\rvert < \varepsilon$.

Theorem (Sequential criterion)**.** $\lim_{x\to a} f(x) = L$ *iff for every sequence* $x_n \to a$ *with* $x_n \neq a$, $f(x_n) \to L$.

Use the sequential criterion to prove non-existence. Find two sequences both tending to a such that $f(x_n) \to L_1$ and $f(y_n) \to L_2 \neq L_1$. Classic: $\lim_{x\to 0} \sin(1/x)$ does not exist — pick $x_n = 1/(n\pi) \to 0$ giving 0, and $y_n = 2/[(4n+1)\pi] \to 0$ giving 1.

3.1 Indeterminate forms

The seven indeterminate forms: $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0\cdot\infty$, $\infty-\infty$, 0^0, ∞^0, 1^∞. Each requires algebraic massaging or L'Hôpital before evaluation.

Reduction tricks.

- $0 \cdot \infty$: rewrite as $0/0$ or ∞/∞.
- $\infty - \infty$: combine over a common denominator, or factor the dominant term.
- 0^0, ∞^0, 1^∞: take ln, get $0 \cdot \infty$, then proceed.

3.2 L'Hôpital's rule (basic and extended)

Theorem (L'Hôpital — 0/0 form)**.** *If* $\lim_{x\to a} f = \lim_{x\to a} g = 0$*,* $g' \neq 0$ *near* a*, and* $\lim \dfrac{f'}{g'} = L$*, then* $\lim \dfrac{f}{g} = L$*.*

Theorem (Extended L'Hôpital — anything/∞)**.** *If* $\lim_{x\to a} g = \infty$ *(the numerator's limit is irrelevant),* $g' \neq 0$ *near* a*, and* $\lim \dfrac{f'}{g'} = L$*, then* $\lim \dfrac{f}{g} = L$*.*

Why "anything". Surprisingly, in the $\frac{\bullet}{\infty}$ case the numerator f need not tend to anything in particular — as long as the ratio of derivatives behaves, the ratio itself does. *Caveat:* L'Hôpital can fail when $\lim \dfrac{f'}{g'}$ doesn't exist, even if $\lim \dfrac{f}{g}$ does. Classic example: $\lim_{x\to\infty} \dfrac{x+\sin x}{x} = 1$ (by squeeze), but $\dfrac{(x+\sin x)'}{x'} = 1 + \cos x$ has no limit — so L'Hôpital is silent here. Always check the derivative-ratio limit exists before concluding.

Common pitfalls.

- Applying L'Hôpital to a form that isn't indeterminate (e.g., $\lim_{x\to 0} \dfrac{x}{1+x}$ is just 0).
- Forgetting that $g' \neq 0$ near a is required.
- Iterating without checking the new form is still indeterminate.

3.3 Hardy's e^x trick

When evaluating limits of the form $\lim_{x\to\infty}$(rational/algebraic) where direct manipulation is awkward, multiply numerator and denominator by e^{-kx} (or e^{kx}) to convert exponential decay into a more tractable form. The same idea (multiplying by a clever exponential) underpins many tricks for limits at infinity, including:

Comparing growth rates. For any $\alpha > 0$ and $\beta > 0$:

$$\ln^\alpha x \ll x^\beta \ll e^{\gamma x} \ll x^x \quad \text{as } x \to \infty.$$

Specifically: $\lim x^\beta / e^{\gamma x} = 0$, $\lim \ln x / x = 0$, $\lim x^n / e^x = 0$ for any n.

The e^{-x} multiplication trick. To show $\lim_{x\to\infty} x^n e^{-x} = 0$ *without* L'Hôpital: write

$$x^n e^{-x} = \frac{x^n}{e^x} = \frac{x^n}{1 + x + \frac{x^2}{2!} + \cdots + \frac{x^{n+1}}{(n+1)!} + \cdots} \leq \frac{x^n}{x^{n+1}/(n+1)!} = \frac{(n+1)!}{x} \to 0.$$

Pick the right term in the Taylor expansion of e^x to dominate the polynomial.

Showcase: $f + f' \to 0$ implies $f \to 0$. A beautiful instance of the e^x trick combined with extended L'Hôpital. Suppose f is differentiable on some $[a, \infty)$ and $\lim_{x\to\infty} \big(f(x) + f'(x)\big) = 0$. We show $\lim_{x\to\infty} f(x) = 0$. Multiply numerator and denominator by e^x:

$$f(x) = \frac{e^x f(x)}{e^x}.$$

The denominator $e^x \to \infty$, so this is the $\frac{\bullet}{\infty}$ form for which extended L'Hôpital applies. Compute the derivative ratio:

$$\frac{(e^x f(x))'}{(e^x)'} = \frac{e^x f(x) + e^x f'(x)}{e^x} = f(x) + f'(x) \to 0.$$

By extended L'Hôpital, $\lim f(x) = 0$. *Why this works:* multiplying by e^x turns the awkward sum $f + f'$ into the derivative of a product (since $(e^x f)' = e^x(f + f')$), and the extended form of L'Hôpital handles the rest — crucially, we never needed to assume f has a limit a priori.

Generalisation. The same trick gives: if $f' + cf \to L$ as $x \to \infty$ for some $c > 0$, then $f \to L/c$. Multiply by e^{cx} instead of e^x.

4 Continuity

4.1 Three equivalent definitions

For f defined on a neighbourhood of a:

- **Limit definition:** f is continuous at a iff $\lim_{x\to a} f(x) = f(a)$.
- **Sequential definition:** f is continuous at a iff for every sequence $x_n \to a$, $f(x_n) \to f(a)$.

- **ε-δ definition:** f is continuous at a iff for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|x - a| < \delta$ implies $|f(x) - f(a)| < \varepsilon$.

These are equivalent for functions defined on $\mathbb{R}$. The sequential definition is usually easiest to disprove continuity (find a bad sequence); the ε-δ definition is needed for uniform continuity.

4.2 Types of discontinuity

At a point x_0 where f is discontinuous (let $L^{\pm} = \lim_{x \to x_0^{\pm}} f(x)$):

- **Removable:** $L^- = L^+ = L$, but $L \neq f(x_0)$. Patchable.
- **Jump:** L^-, L^+ both exist, $L^- \neq L^+$.
- **Essential (oscillatory or infinite):** at least one one-sided limit fails to exist.

Removable and jump discontinuities are called *first-kind* (or *simple*); essential ones are *second-kind.*

4.3 Major theorems on continuous functions

Theorem (Intermediate Value Theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous and y lies between $f(a)$ and $f(b)$, then $f(c) = y$ for some $c \in [a, b]$.*

Use. The classical existence engine: roots from sign changes, fixed points via $h(x) = f(x) - x$, “meeting” problems via $h(x) = f(x) - g(x)$.

Theorem (Extreme Value Theorem (EVT))**.** *If $f : [a, b] \to \mathbb{R}$ is continuous on a closed bounded interval, then f attains its maximum and minimum on $[a, b]$: there exist $c, d \in [a, b]$ with $f(c) \leq f(x) \leq f(d)$ for all $x \in [a, b]$.*

All three hypotheses matter.

- *Continuity* is essential: $f(x) = 1/x$ on $(0, 1]$ extended by $f(0) = 0$ is bounded but unbounded near 0 when continuity fails.
- *Closed* interval is essential: $f(x) = x$ on $(0, 1)$ has no max or min — the supremum and infimum are not attained.
- *Bounded* interval is essential: $f(x) = x$ on $[0, \infty)$ attains its min at 0 but has no max.

Why it matters. EVT guarantees that optimisation problems on closed bounded intervals *have* a solution — the bedrock under every Lagrange-multiplier or critical-point analysis. Combined with Fermat’s stationary-point theorem, EVT yields the standard recipe: to maximise a continuous f on $[a, b]$, evaluate f at all critical points in (a, b) and at the endpoints a, b, then pick the largest. EVT also powers the proof of Rolle’s theorem (interior extremum exists $\Rightarrow f' = 0$ there).

Other essentials.

- **Boundedness:** A continuous function on a closed bounded interval is bounded. (Immediate from EVT.)
- **Uniform continuity:** A continuous function on a closed bounded interval is uniformly continuous (Heine–Cantor).
- **Composition:** The composition of continuous functions is continuous.

4.4 Pathological-continuity facts

Continuity sets are G_δ. For any function $f : \mathbb{R} \to \mathbb{R}$, the set of points where f is continuous is a G_δ set (countable intersection of open sets). Consequence: *no function can be continuous exactly on* $\mathbb{Q}$, since $\mathbb{Q}$ is not G_δ in $\mathbb{R}$. But continuity *exactly on the irrationals* is possible (Thomae’s function).

Continuous + injective on an interval $\Rightarrow$ monotone. A standard MCQ trap: if $f : [a, b] \to \mathbb{R}$ is continuous and injective, then f is strictly monotone (and its inverse is also continuous).

Density of points of continuity. If f has the IVP and is monotone on every subinterval... etc. (Many such results follow from G_δ structure plus Baire’s theorem.)

5 Differentiability

Theorem (Definition)**.** *f is differentiable at a if $\lim_{h \to 0} \dfrac{f(a + h) - f(a)}{h}$ exists. The limit is denoted $f'(a)$.*

5.1 Differentiability vs continuity

Differentiable $\Rightarrow$ continuous. If $f'(a)$ exists, then $\lim_{x \to a} f(x) = f(a)$, because

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a) \to f'(a) \cdot 0 = 0.$$

Continuous $\not\Rightarrow$ differentiable. The function $f(x) = |x|$ is continuous everywhere, but not differentiable at 0. Even more striking: there exist functions continuous everywhere but differentiable nowhere (Weierstrass's function).

5.2 What derivatives can and cannot look like

Theorem (Darboux)**.** *If f is differentiable on $[a, b]$, then f' has the intermediate-value property.*

Consequence. Even if f' is discontinuous, the only allowed discontinuities are *essential* (oscillatory) — jumps and removable discontinuities are forbidden. The Heaviside step is therefore not the derivative of *any* function.

Derivative-limit theorem. If f is continuous on a neighbourhood of a and $\lim_{x\to a} f'(x) = L$ exists, then f is differentiable at a with $f'(a) = L$. So if a derivative has a limit at a point, that limit *is* the derivative there. (This is essentially Darboux + MVT.)

5.3 Mean value theorems (essentials)

- **Rolle:** $f(a) = f(b)$, differentiable $\Rightarrow f'(c) = 0$ for some interior c.
- **Lagrange (MVT):** $f'(c) = \frac{f(b)-f(a)}{b-a}$ for some interior c.
- **Cauchy:** $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$ for some interior c, when $g' \neq 0$.

6 Convexity and concavity

Theorem (Definition)**.** *f is* convex *on an interval I if for all $x, y \in I$ and $\lambda \in [0, 1]$:*

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y).$$

f is concave *if the reverse inequality holds (equivalently, $-f$ is convex).* Strict *convexity if the inequality is strict for $\lambda \in (0, 1)$ and $x \neq y$.*

6.1 Tests for convexity

- **First-derivative test:** f convex on I iff f' is non-decreasing on I.
- **Second-derivative test:** If f is twice differentiable, f convex iff $f'' \geq 0$, strictly convex if $f'' > 0$ (the converse needs care: strict convexity allows f'' to vanish at isolated points, e.g., x^4).
- **Tangent-line characterisation:** f convex iff its graph lies above every tangent: $f(x) \geq f(a) + f'(a)(x-a)$ for all x, a.
- **Chord characterisation:** f convex iff its graph lies below every chord.

6.2 Inflection points

A point where f changes from convex to concave (or vice versa). At a smooth inflection, $f''(x_0) = 0$ *and* f'' changes sign there. Note: $f''(x_0) = 0$ alone is not sufficient (e.g., $f(x) = x^4$ has $f''(0) = 0$ but no inflection).

6.3 Jensen's inequality

For convex f and weights $\lambda_i \geq 0$ with $\sum \lambda_i = 1$:

$$f\Big(\sum \lambda_i x_i\Big) \leq \sum \lambda_i f(x_i).$$

Reverse for concave f. Specialisations include AM–GM (apply to $f(x) = -\ln x$, concave), Cauchy–Schwarz, and many olympiad inequalities.

6.4 Tangent-line and chord inequalities

- $e^x \geq 1 + x$, equality at $x = 0$ (tangent to e^x at 0; convexity).
- $\ln(1 + x) \leq x$ for $x > -1$, equality at 0 (tangent; ln is concave).
- $\sin x \leq x$ for $x \geq 0$, equality at 0 (concavity of sin on $[0, \pi]$).
- For $0 < a < b$: $\dfrac{b-a}{b} < \ln \dfrac{b}{a} < \dfrac{b-a}{a}$ (chord/MVT for ln).

7 Quick-reference cheat sheet

Pattern in the problem	Tool / trick to try first
$\sin n/n$, oscillating bounded $\times$ small	**Squeeze**
$\sum_{k=1}^{n} f(k)/g(n)$ ratio with sum on top	**Stolz–Cesàro**
Anything with $n!$ in nth-root limits	**Stirling**: $n! \sim (n/e)^n\sqrt{2\pi n}$
$\binom{kn}{n}^{1/n}$ or $\binom{n}{\alpha n}^{1/n}$	**Stirling** on each factorial
$a_{n+1}/a_n \to L > 0$, want $a_n^{1/n}$	Limit equals L (Cesàro on logs)
$\sum 1/n^p$	Converges iff $p > 1$
$\sum$ with r^n growth	**Geometric** or **ratio test**
$\sum$ with no obvious closed form	**Comparison** or **integral test**
Alternating $\sum(-1)^n a_n$, $a_n \downarrow 0$	**Leibniz** (conditional convergence)
$\frac{0}{0}$ or $\frac{\infty}{\infty}$ in function limits	**L'Hôpital** (check derivatives have a limit!)
$0 \cdot \infty$, $\infty - \infty$	Algebra to $\frac{0}{0}$ or $\frac{\infty}{\infty}$
0^0, ∞^0, 1^∞	Take ln first
x^n/e^x as $x \to \infty$	Use e^x Taylor expansion (Hardy trick)
Show lim does not exist	Two sequences with different image-limits
Function continuous on $[a, b]$ + injective	It's monotone
"Continuous exactly on $\mathbb{Q}$"	Impossible ($\mathbb{Q}$ not G_δ)
"Continuous exactly on $\mathbb{R} \setminus \mathbb{Q}$"	Possible (Thomae)
Show $f' \geq 0 \Rightarrow f$ increasing	**MVT**
f has IVP but discontinuous	Discontinuities must be essential
Derivative jump claimed	Impossible (Darboux)
Inequalities involving $e^x, \ln, \sin, \arctan$	Tangent-line / convexity / MVT
Jensen-type sums of $f(x_i)$	**Convexity** of f

Three universal moves: (1) When in doubt about a function-limit, try the sequential criterion. (2) For sequence ratios with sums, Stolz–Cesàro often beats algebra. (3) For factorials in nth roots, Stirling is almost always the fastest path.